Least Squares Fit: $\hat{y} = m\hat{x} + b$

Given the following three points, find the line that fits them: $(-4,-2), (2,-8), (3,-7)$.

Mean: $n=3,\ x=\{-4,2,3\},\ y=\{-2,-8,-7\}$

$\bar{x} = \frac{1}{n}\sum x_i$	$\bar{x} = \frac{1}{3}(-4+2+3) = \frac{1}{3}$
$\bar{y} = \frac{1}{n}\sum y_i$	$\bar{y} = \frac{1}{3}(-2-8-7) = -\frac{17}{3}$

Deviations:

$x_i - \bar{x}$	$y_i - \bar{y}$
$x_1 - \bar{x} = -4 - \frac{1}{3} = \frac{-12-1}{3} = -\frac{13}{3}$	$y_1 - \bar{y} = -2 - (-\frac{17}{3}) = \frac{-6+17}{3} = \frac{11}{3}$
$x_2 - \bar{x} = 2 - \frac{1}{3} = \frac{6-1}{3} = \frac{5}{3}$	$y_2 - \bar{y} = -8 - (-\frac{17}{3}) = \frac{-24+17}{3} = -\frac{7}{3}$
$x_3 - \bar{x} = 3 - \frac{1}{3} = \frac{9-1}{3} = \frac{8}{3}$	$y_3 - \bar{y} = -7 - (-\frac{17}{3}) = \frac{-21+17}{3} = -\frac{4}{3}$

Sum of Squares:

$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
$(x_1-\bar{x})^2 = (-\frac{13}{3})^2 = \frac{169}{9}$	$(y_1-\bar{y})^2 = (\frac{11}{3})^2 = \frac{121}{9}$
$(x_2-\bar{x})^2 = (\frac{5}{3})^2 = \frac{25}{9}$	$(y_2-\bar{y})^2 = (-\frac{7}{3})^2 = \frac{49}{9}$
$(x_3-\bar{x})^2 = (-\frac{8}{3})^2 = \frac{64}{9}$	$(y_3-\bar{y})^2 = (-\frac{4}{3})^2 = \frac{16}{9}$
$\sum (x_i-\bar{x})^2 = \frac{169}{9}+\frac{25}{9}+\frac{64}{9} = \frac{258}{9}$	$\sum (y_i-\bar{y})^2 = \frac{121}{9}+\frac{49}{9}+\frac{16}{9} = \frac{186}{9}$

Sum of Cross Deviations: $(x_i - \bar{x})(y_i - \bar{y})$

$(x_1-\bar{x})(y_1-\bar{y}) = (-\frac{13}{3})(\frac{11}{3}) = -\frac{143}{9}$
$(x_2-\bar{x})(y_2-\bar{y}) = (\frac{5}{3})(-\frac{7}{3}) = -\frac{35}{9}$
$(x_3-\bar{x})(y_3-\bar{y}) = (\frac{8}{3})(-\frac{4}{3}) = -\frac{32}{9}$
$\sum (x_i-\bar{x})(y_i-\bar{y}) = (-\frac{143}{9})+(-\frac{35}{9})+(-\frac{32}{9}) = -\frac{210}{9}$

Least Squares Fit: $\hat{y} = m\hat{x} + b$

Slope:

$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{-210/9}{258/9} = \frac{-210}{9} \div \frac{258}{9} = \frac{-210}{9} \cdot \frac{9}{258} = \frac{-210}{258} = \frac{-35}{43}$

Intercept:

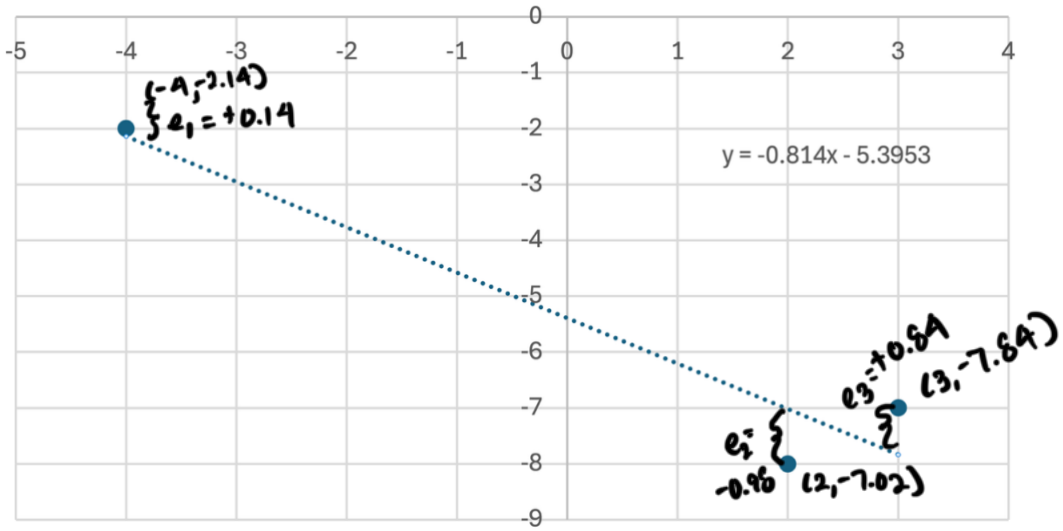
$b = \bar{y} - m\bar{x} = \left(\frac{-17}{3}\right) - \left(\frac{-35}{43}\right)\left(\frac{1}{3}\right) = \frac{-17}{3} + \frac{35}{129} = \frac{-731+35}{129} = \frac{-696}{129} = \frac{-232}{43}$

Final Line:

$\hat{y} = m\hat{x} + b = \frac{-35}{43}\hat{x} - \frac{232}{43} \Rightarrow \hat{y} = -0.81\hat{x} - 5.4$

Predictions:

x_i	y_i	$\hat{y}_i = m\hat{x}_i + b$	$(\hat{x}_i, \hat{y}_i)$	$Residuals: e_i = y_i - \hat{y}_i$
-4	-2	$\frac{-35}{43}(-4) - \frac{232}{43} = \frac{-92}{43} = -2.14$	$(-4, -2.14)$	$-2 - (-2.14) = +0.14$
2	-8	$\frac{-35}{43}(2) - \frac{232}{43} = \frac{-302}{43} = -7.02$	$(2, -7.02)$	$-8 - (-7.02) = -0.98$
3	-7	$\frac{-35}{43}(3) - \frac{232}{43} = \frac{-337}{43} = -7.84$	$(3, -7.84)$	$-7 - (-7.84) = +0.84$



Mean Square Error:

$\frac{1}{n}\sum(y_i - \hat{y}_i)^2 = \frac{1}{3}\left[0.14^2 + (-0.98)^2 + 0.84^2\right] = \frac{1}{3}(0.0196 + 0.9604 + 0.7056) = \frac{1.6856}{3} = 0.56$

Interpretation:

MSE measures how far our predictions are from the actual values on average. We want the MSE to be as small as possible for the best fit line and most accurate predictions. Larger datasets require ML because it is impossible to do manually.